Biomechanical Comparison of Forward and Lateral Lunges

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Abstract & Overview

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1. Biomechanics of the Lunge

The lunge is a unilateral compound lower-body movement targeting the quadriceps, gluteus maximus, hamstrings, and calves while challenging core and pelvic lateral stability.
Key joint angles:
- Standing / Up Phase: Front and rear legs extended at 160° to 180°.
- Bottom Lunge Phase: Front knee bends to ≤ 100° (ideally 90° tibiofemoral angle). Rear knee lowers directly toward the floor to approximately 90° without crashing into the mat.
- Torso Alignment: Torso remains upright (torso angle ≥ 160° relative to vertical).

2. Common Form Errors

• Error A: Lateral Loss of Balance ("OFF BALANCE")
  - Biomechanical Cause: Excessive lateral deviation of the center of mass (offset > 0.10 normalized units between shoulder midpoint and hip midpoint).
  - Root Causes: Stepping in a straight line ("tightrope walking") rather than stepping along parallel tracks (railroad tracks). Weak hip abductors (gluteus medius).
  - Correction: Step forward while maintaining a hip-width base of support. Look forward and engage core stabilizers.
• Error B: Front Knee Over-Extending Forward of the Toes
  - Biomechanical Cause: Excessive forward tibial shear causing high patellofemoral compressive force.
  - Correction: Step farther forward and direct descent vertically downward rather than driving forward.
• Error C: Torso Leaning Forward
  - Correction: Keep chest high, shoulders down and back, and core braced.

3. Recommended Camera Setup for Lunges

- Angle: Side (sagittal) view or 45-degree angle.
- Distance: 2.5 to 3.5 meters.
- Framing: Full body framing showing both feet on the floor and full torso upright.